

TER starting kit

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1 Tools

GitHub repository: https://github.com/CTInfer/Finetuning_few_shot

LabIA documentation: <https://doc.lab-ia.fr/>

2 Sub-Subjects

2.1 Finetuning on NLI4PR

The aim of this sub-subject is to finetune pretrained LLMs using the NLI4PR dataset. Naive and more sophisticated finetuning approaches will be used. You'll compare your results with the ones obtained with the sub-subject finetuning on NLI4CT. If you both result into well performing models, we could try to using model merging or aggregating both datasets for training.

2.2 Finetuning on NLI4CT

The aim of this sub-subject is to finetune pretrained LLMs using the NLI4CT dataset. Naive and more sophisticated finetuning approaches will be used. You'll compare your results with those obtained using the sub-subject finetuning on NLI4PR. If both result in well-performing models, we could try using model merging or aggregating both datasets for training.

2.3 Few-shot learning

The aim of this sub-subject is to test several approaches for selecting optimal examples to perform Few-shot learning, both using NLI4CT and NLI4PR. Previous approaches such as KATE (Liu et al., 2021) can be explored. We can module different variations of parameters: the number of examples provided in the prompt, how to select them, how to present/format them ...

3 Tasks

Will be adjusted in function of your progress.

The main tasks are the following:

1. Literature review and definition of approaches to address the problems.
2. Training and evaluation of LLMs on the NLI4CT and NLI4PR tasks using different finetuning techniques.
3. Experimenting using different optimization techniques for Few-shot learning, using NLI4CT and NLI4PR.
4. Comparison with the state of the art and qualitative analysis of the results obtained.

4 Literature

To get to know the NLI task:

- Bowman et al. (2015)
- Chen et al. (2017)
- Huertas-Tato et al. (2022)

Applications of NLI in the clinical domain:

- Jullien et al. (2023a)
- Jullien et al. (2023b)
- Percha et al. (2021)

4.1 Datasets

- Multi-genre NLI: MNLI (Williams et al., 2018)
- Medical NLI: MedNLI (Shivade, 2017)
- NLI for Clinical Trials: NLI4CT (Jullien et al., 2023a)
- Natural Language Inference for Clinical Trial Patient Recruitment: NLI4PR Aguiar et al. (2025)

4.2 Baselines

- Really popular baseline in NLI: ESIM(Chen et al., 2017)
- A T5 trained on scientific texts: SciFive (Phan et al., 2021)
- A new multilingual and biomedical LLM BioMistral (Labrak et al., 2024)
- See models used in the given literature ...

5 Miscellaneous

NLP tutorials:

A collection of example scripts and notebooks to get used to the HuggingFace API and how to train models, load datasets, etc.: Notebooks and scripts. Have a look at the Sequence Classification ones in particular. The NLP course by HuggingFace, especially the LLM finetuning chapter.

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